Department of Computer Science & Engineering
The University of Texas at Arlington

System Requirements Specification
CSE 4316: Senior Design I
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IGVC
MavTruck

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Revision History

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1 Product Concept

This section describes the purpose, use, and intended audience for the MavTruck product.The MavTruck product is a system that will be able to autonomously guide its way through a series of obstacles on a course. This competition and course will be held by IGVC.

1.1 Purpose and Use

This product will be used in the Intelligent Ground vehicle Competition. It will need to autonomously navigate its way through the course created. Although the main goal is to qualify for and win the competition, we will achieve other goals along the way. We aim to improve our understanding of robotics, which includes motor control, sensors, and computer vision. Additionally, while working as a team, we will improve our skill set of collaboration, innovation, and composition

1.2 Intended Audience

Our product for this competition is a little different than an actual product that will be used/sold to customers, but there are still many intended audiences. The main audience are the participants. They will focus on development, innovation, and problem solving. They will be able to assess their product as well as other teams' products as well. Another main audience would be the judges. They will be the ones deciding if the products meet the requirements to attempt to qualify and judging the products on whether they actually qualify for the competition. If we qualify, the judges will also decide the winner of the competition, by accuracy moving through the course and the speed at which is maintained. The last audience will be the general public. This can include sponsors/industry companies, who will be recruiting potential talent, branding, and networking, educators, who will be interested in the outcomes of the teams they may be leading, and the media, who may be covering tech innovations.

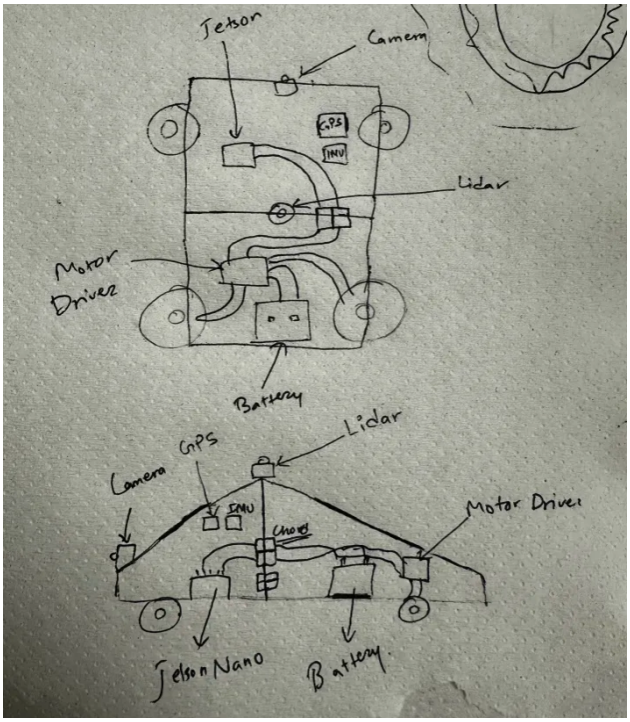

Figure 1: MavTruck conceptual drawing

2 Product Description

This section provides the reader with an overview of MavTruck. The primary operational aspects of the product, from the prospective of end users, maintainers and administrators, are defined here. The key features and functions found in the product, as well as critical user interactions and user interfaces are described in detail.

2.1 Features & Functions

Our product will be able to detect different objects and lane lines to make correct decisions and guide through the course. We will utilize many different components to make this happen. The first component is the camera. The camera will be used to detect the lane lines to know where to guide the vehicle. It will also detect the shape and color of objects to know which decision to make as well. The next component is LiDAR. The LiDAR will also be used as detection for objects in every direction. In addition to that, we will utilize an IMU and GPS to know the location and speed/acceleration of our vehicle. We will utilize motor drivers to control the back wheels. The final and most important component of our vehicle is the Jetson Nano. This device that will take in all the inputs and based on them, it will control the motor drivers to determine the speed and direction the vehicle will go.

2.2 External Inputs & Outputs

The inputs of our product will come from the environment around it. The competition has released many different tests our product must pass in order to qualify and complete the course. There are about 21 tests given that could be potential inputs for our device. This includes lane detection, pothole detection, pedestrian detection, stop sign detection, parking detection, and many more. With each input, there will be an output and each output will be different. The main outputs of our product will be the motor drivers. This output will control the direction and speed our product will go. For example, if there is a pedestrian in the middle of the lane lines, our system must detect it and come to a complete stop within 5 to 7 feet of them. The inputs came from the camera and LiDAR and the outputs came from the IMU and the motor drivers.

2.3 Product Interfaces

The only images we can showcase are the vehicle tests published in the IGVC 2026 rules. Each test shows the potential inputs and what the expected outputs must be. This is Test FI.1 on page 27 of the official IGVC 2026 rules. The test is called "Static Pedestrian Detection." It shows a pedestrian in the lane lines, explains how the vehicle must react and the evaluations at the end.

Test FI.1 Static Pedestrian Detection

1. Test Goal

This test is intended to evaluate detection of a mannequin using traditional Machine Vision algorithms. A mannequin wears ORANGE construction vest. A GUI interface with extracted orange blob MUST be present during a run. There are NO PENALTIES for crossing or moving over a line.

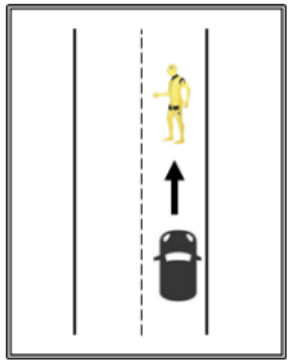

Figure 7: Machine Vision Tests. Static Pedestrian Detection

2. Test Setup

The following items shall be placed on the road:

- **Barrel 1** to indicate starting point at which vehicle is stationary

3. Test Script
   1. Begin test run
   2. Judge pushes 'start' button
   3. The extracted orange blob is present on the screen.
   4. End test run

4. Evaluation
   Fail Criteria – no GUI interface is present during the run, incorrect identification of the shape/object
   Penalties – no penalties for crossing or moving over the lines, in case if vehicle is moving during the test

Figure 2: Test FI.1 from IGVC Official 2026 Rules

3 Customer Requirements

We are developing a modular computer vision system with ROS2 using OPENCV for the UTA IGVC vehicle, based on the requirements of Professor Christopher Conly and the IGVC rules. The system must integrate with the current vehicle architecture and will be trained using a custom data set of obstacles and scenarios expected at the competition.

3.1 COMPUTER VISION AND FUSION SYSTEM

3.1.1 Description

The vehicle will use a forward-facing camera for lane detection and an RPLIDAR S3 sensor for obstacle detection. The Jetson Orin Nano will process these inputs using custom Python code (OpenCV for image processing and Python-based LiDAR filtering). GPS and IMU data are used for position and heading estimation. These systems together enable the vehicle to remain within lane boundaries, avoid obstacles, and navigate the IGVC course.

3.1.2 Source

The IGVC 2026 rules and requirements.

3.1.3 Constraints

- Performance is limited
- LIDAR accuracy depends on environment reflectivity and weather.
- GPS accuracy varies in outdoor conditions.

3.1.4 Standards

IGVC competition rules requiring lane following, obstacle avoidance, and safe autonomous navigation.

3.1.5 Priority

- Critical

3.2 SENSOR TESTING AND VALIDATION DATA

3.2.1 Description

The team will perform testing of the camera, LiDAR, IMU, and GPS in controlled environments to validate lane detection accuracy, obstacle detection performance, and localization reliability. Collected sensor data (camera frames, LiDAR scans, GPS logs) will be analyzed offline to tune thresholds and improve the perception pipeline.

3.2.2 Source

Team testing procedures based on IGVC rules.

3.2.3 Constraints

Weather, lighting conditions, and limited access to outdoor test spaces may restrict data collection.

3.2.4 Standards

Sensor testing follows IGVC guidelines requiring detection of lane lines, obstacles, and boundary markers.

3.2.5 Priority

- High

4 Packaging Requirements

The IGVC robot will be transported in separate modular components to ensure safe delivery and efficient on-site assembly. Each subsystem-including the drive base, vision system, and computing unit-must be securely packaged to prevent damage during shipping and must support quick reassembly using minimal tools. Proper packaging is critical for maintaining functionality, ensuring compliance with competition deadlines, and protecting delicate hardware from environmental hazards.

4.1 Modular Shipping and Assembly Design

4.1.1 Description

The IGVC vehicle must be packaged in a robust, modular frame that can be easily disassembled for transport and reassembled for testing and competition. It must safely enclose all electronic and mechanical components including sensors, computing units, and batteries. Weatherproofing and shock absorption should be incorporated to protect sensitive components.

4.1.2 Source

IGVC rules and logistics requirements

4.1.3 Constraints

The design should reduce the overall footprint of the packaged unit. The base must be assembled with minimal tools and effort. All parts must be secured to resist shifting or damage during handling. Packaging should protect the hardware from moisture, dust, and shock.

4.1.4 Standards

ASTM D7386 a Performance testing protocols for shipping containers.

4.1.5 Priority

Critical - If the drive system is damaged or delayed in shipping, the robot cannot be fielded.

5 Performance Requirements

The autonomous vehicle must operate within a defined speed range during competition events. Maintaining this range is essential for both safety and regulatory compliance. Any run that violates these speed constraints is disqualified according to IGVC rules.

5.1 Autonomous Speed Compliance

5.1.1 Description

The robot must maintain a speed that complies with IGVC competition rules during fully autonomous operation. Speeds outside of this range will result in penalties or disqualification. The system must be configured such that maximum speed limits are enforced by hardware, and cannot be adjusted after passing qualification.

5.1.2 Source

IGVC rules and logistics requirements

5.1.3 Constraints

Speed limits must be enforced via onboard hardware and remain unchanged after initial qualification. Minimum Speed: The robot must sustain a minimum speed of 1 mile per hour over a 44-foot section. Maximum Speed: The robot must not exceed 5 miles per hour at any time.

5.1.4 Standards

Speed thresholds and safety requirements as outlined by IGVC competition rules and inspection guidelines.

5.1.5 Priority

Critical - Violation of speed constraints invalidates performance runs and may lead to disqualification.

6 Safety Requirements

To meet the IGVC safety compliance standards, our autonomous ground vehicle will incorporate a set of essential features designed to ensure safe operation during testing and competition.

The vehicle will be equipped with two independent emergency stop systems. To ensure the system complies with expected standards and minimizes risk during operation, both manual and remote emergency stop features will be implemented. The physical Emergency Stop will allow any nearby user to shut off vehicle power instantly, while the wireless option ensures remote shutdown capabilities during autonomous operation and testing.

Additionally, the vehicle will feature LED indicators to visually convey its operating status. These features are designed to minimize risks, enable quick response during emergencies, and communicate the vehicle's operational state to nearby teams. As our goal is to build a safe, responsive system that reduces risk to both people and hardware during all phases of operation

6.1 Laboratory equipment lockout/tagout (LOTO) procedures

6.1.1 Description

To ensure safe handling of electrical and mechanical components during the development of the IGVC vehicle, our team will follow a structured LOTO protocol outlined by OSHA. Any equipment posing a potential hazard during maintenance or configuration, such as power tools, motor controllers, or high-current supplies, will be secured with physical locks and tagged to indicate their status.

6.1.2 Source

CSE Senior Design laboratory policy

6.1.3 Constraints

Equipment usage, due to lock removal policies, will be limited to availability of the course instructor and designed teaching assistants.

6.1.4 Standards

Occupational Safety and Health Standards 1910.147 - The control of hazardous energy (lockout/tagout).

6.1.5 Priority

Critical

6.2 National Electric Code (NEC) wiring compliance

6.2.1 Description

To ensure safety and reliability, all electrical systems in our IGVC vehicle will be designed and assembled in accordance with the National Electrical Code (NEC) standards. This includes wire runs, insulation, grounding, enclosures, over-current protection, and all other specifications.

During the wiring process, team members will verify compliance with NEC guidelines for voltage handling, connector types, and enclosure safety. These are intended to prevent electrical hazards, improve system durability, and support safe operation during testing and competition.

6.2.2 Source

CSE Senior Design laboratory policy

6.2.3 Constraints

To comply with NEC standards while ensuring system reliability and safety, our team is following a set of practical design constraints.We're using wire gauges that match the expected current loads and ensuring all high-voltage areas are well insulated and clearly marked. Components will be placed with enough space to avoid accidental contact, especially around connectors or power lines.

We'll also follow proper grounding practices to minimize the risk of electric shock or damage during a fault. A consistent grounding strategy will be implemented to provide safe fault current paths. Electrical components will be installed in secure, dry locations, and the system will include circuit protection, such as fuses or breakers, to guard against short circuits and overloads.

6.2.4 Standards

NFPA 70

6.2.5 Priority

Critical

6.3 RIA robotic manipulator safety standards

6.3.1 Description

Not applicable. The IGVC vehicle does not include any robotic arms or manipulators, nor are such components planned for current or future development.

6.3.2 Source

Not applicable.

6.3.3 Constraints

Not applicable.

6.3.4 Standards

Not applicable.

6.3.5 Priority

Low

7 Maintenance & Support Requirements

This section outlines what is needed to maintain the IGVC vehicle after deployment. It covers hardware and software components that must be easy to service, replace, or update in the field. Proper labeling, documentation, and modular design are essential to ensure the system remains functional throughout testing and competition use.

7.1 Maintainance and Supoprt

7.1.1 Description

The vehicle consists of a base frame (breakable for transport), camera, LiDAR, GPS, Jetson Nano, emergency stop system, motor driver, and wheelchair motors. Each component must be modular and serviceable. The breakable frame must be robust and designed for repeated assembly/disassembly. Wiring for sensors and motors must be labeled and easily traceable for quick replacement.

7.1.2 Source

IGVC computer vision team.

7.1.3 Constraints

Maintenance must be doable in the field with basic tools. Frame joints must be secure during operation but easy to separate for travel. No soldering should be required for swapping components.

7.1.4 Standards

Best practices for field-deployable robotics (e.g., modular connector use, wiring harnesses); follow National Electric Code where applicable.

7.1.5 Priority

Hight

8 Other Requirements

This section lists additional requirements needed to complete the IGVC system. It includes software compatibility, cybersecurity, and modularity considerations that do not fit under other categories but are essential for full system functionality, portability, and reliability during competition.

8.1 Software and Firmware support

8.1.1 Description

The Jetson Orin Nano will run custom Python scripts for lane detection, LiDAR obstacle processing, and high-level motor command generation. Firmware on the TM4C123GXL microcontroller handles low-level control tasks including PWM motor control, encoder counting, and emergency stop logic.

8.1.2 Source

IGVC software and integration team.

8.1.3 Constraints

Python 3.10+ runtime, GPU-enabled OpenCV on Jetson, TivaWare firmware environment for TM4C.

8.1.4 Standards

Must be updatable via SSH or direct USB. Firmware for motor controllers must be documented and reloadable if corrupted.

8.1.5 Priority

Hight

8.2 Cross-Platform Development

8.2.1 Description

Sensor processing code (camera, LiDAR, GPS parsing) should run on both a desktop Ubuntu system for development and on the Jetson Orin Nano for deployment.

8.2.2 Source

Vision and integration development pipeline.

8.2.3 Constraints

OpenCV and Python library versions must match across platforms. Jetson platform requires hardware-accelerated builds.

8.2.4 Standards

Python 3.10+, OpenCV 4.x.

8.2.5 Priority

Hight

9 Future Items

In this section, we list requirements with a priority of "Future". These requirements were considered important and discussed but will not be included in the current prototype due to limitations in time, budget, or technical feasibility.

9.1 Hydraulic Hopping Mechanism

9.1.1 Description

We propose adding a hydraulic hopping mechanism to the IGVC vehicle to allow vertical movement and "hopping" behavior, inspired by lowrider car suspension systems. The goal is to demonstrate advanced motion capabilities and increase crowd engagement during demonstrations or competitions. The system would use hydraulic or pneumatic actuators placed on all four corners of the chassis to provide controlled lift and bounce, enabling the vehicle to hop vertically or tilt in different directions.

9.1.2 Source

Team concept proposal and mechanical design inspiration from automotive lowrider systems.

9.1.3 Constraints

Due to limited budget, time, and mechanical resources, implementing a fully functional hydraulic or pneumatic system is not feasible in the current development cycle. Weight, safety, and integration with existing components also present challenges. The team is also composed of computer engineers only which brings a big challenge on adding mechanical components.

9.1.4 Standards

SAE J1349 Hydraulic Components, ISO 4413 Hydraulic Fluid Power Safety, and IGVC competition design constraints regarding vehicle motion and stability.

9.1.5 Priority

Future

References

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